

Anabella F. Phelps

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Education

Carnegie Mellon University, Mellon College of Science – Pittsburgh, PA

Expected May 2029

B.S in Computational Biology

3.91/4.00

Related Coursework: Principles of Imperative Computation, Functional Programming, Intro to Quantitative Genetic Analysis, Discrete Mathematics, Organic Chemistry, Probability Theory, Cancer Immunology

Technical Skills

Programming Languages: Python, R, C/C++, Java

Bioinformatics: Seurat, Scanpy, AnnData, Salmon, FastQC, MultiQC, Visium, scRNA-seq/ snRNA-seq

Analysis: Differential expression, cell-type annotation, PCA/UMAP, pathway enrichment, ssGSEA

Tools: Git, Jupyter, RStudio, pandas, NumPy, SciPy, Conda, GEO

Experimental: QuPath, flow cytometry, ELISA, cell culture, RNA isolation, PCR/qPCR, immunohistochemistry

Professional Experience

UPMC Hillman Cancer Center - Systems Immunology Intern - Pittsburgh, PA

August 2025 - Present

- Built a multi-study PDAC single-cell atlas spanning 9 scRNA-seq datasets, harmonizing cell annotations and clinical metadata to enable cross-cohort analysis of tumor-immune-stromal interactions.
- Characterized CAF heterogeneity using differential expression, subtype signatures, dimensionality reduction, and pathway enrichment/ssGSEA to prioritize populations and transcriptional programs for downstream study.
- Engineered reusable R and Python pipelines for scRNA-seq and bulk RNA-seq, integrating QC, normalization, annotation, differential expression, pathway analysis, and visualization for ongoing lab studies.
- Investigated CD37-TLS associations across bulk RNA-seq and spatial transcriptomic cohorts using patient- and region-level statistical analyses to generate hypotheses for downstream study.

Project Experience

AI Literacy for K-5 – CREATE Lab

February - May 2026

- Collaborated on a 6-person interdisciplinary team to develop a 12-module K–5 AI literacy curriculum covering AI decision-making, data sorting, creativity, empathy, and critical evaluation of AI-generated outputs.
- Presented the curriculum at the 2026 Nearby Data Workshop to 150 K–12 educators from multiple states, facilitating hands-on discussion and incorporating educator feedback to improve classroom applicability.

Tertiary Lymphoid Structure Simulator – Term Project

May 2026

- Built an interactive Python simulation of TLS development, implementing immune-cell interactions, state progression, and BFS-based spatial clustering to model progression toward mature germinal-center structures

Quantra - TartanHacks

February 2026

- Collaborated on a 4-person team to develop a financial analytics platform integrating transaction visualization, scenario modeling, and conversational data analysis.
- Engineered a multi-level cash-flow explorer and context-aware chat interface for analyzing spending patterns, transactions, and financial scenarios.

Investigating Vaccinia Virus on Formation of Tertiary Lymphoid Structure in Mouse Models - UPMC

June - August 2025

- Evaluated Vaccinia virus-associated TLS formation in lung cancer using syngeneic mouse models, multispectral imaging, immunohistochemistry, and spatial transcriptomic data.
- Developed a QuPath image-analysis workflow to identify and quantify CD19+ B-cell aggregates across tumor tissue samples, improving spatial TLS assessment.

Spatial Analysis of CD200 in High Grade Serous Ovarian Cancer - UPMC

June - August 2024

- Performed immunohistochemistry across 52 ovarian cancer tissue slides and quantified spatial CD200 expression using Keyence imaging and QuPath analysis.
- Identified elevated CD200-positive cell abundance within ovarian tumor tissue, generating preliminary spatial evidence supporting further investigation of the CD200/CD200R immunoregulatory axis.

Leadership & Honors

Teaching Assistant, 15-112: Fundamentals of Programming and Computer Science | Fall 2026

Youth Enjoy Science Student Mentor, Hillman Cancer Academy | 2025–2026

Pittsburgh Regional Science & Engineering Fair — First Place (Biotechnology) | 2025

